

The Law of Large Numbers in Complex Systems: From Statistical Convergence to Compressed Models

Boris Kriger^{1,2}

¹Information Physics Institute, Gosport, Hampshire, United Kingdom,
`boris.kriger@informationphysicsinstitute.net`

²Institute of Integrative and Interdisciplinary Research, Toronto, Canada,
`boriskruger@interdisciplinary-institute.org`

Abstract

The law of large numbers (LLN) guarantees that sample averages converge to expected values under conditions of independence, identical distribution, and finite expectation. This paper demonstrates that complex adaptive systems—biological, cognitive, and social—systematically violate these conditions, and that this violation is not incidental but structurally necessary for persistence.

We introduce the concept of *agent replaceability* $\mathcal{R}(a)$ as a formally defined measure of the degree to which a system’s structural contribution can be recovered after loss, and prove that decreasing replaceability forces a transition from population-level statistical averaging (the “LLN-strategy”) to individual-level internal modeling (the “model-strategy”). The transitional case of bacterial chemotaxis is analyzed as a system exhibiting both strategies simultaneously.

We show that cognitive biases, heuristic reasoning, and cultural filtering—conventionally treated as deviations from rational inference—are structural consequences of operating in regimes where the LLN is inapplicable, and represent optimal adaptations under the constraint of irreplaceable agents. A scale-dependent compressibility criterion based on Kolmogorov complexity provides a non-circular characterization of this optimality: biases exploit compressible environmental structure at the agent’s operational scale that statistical averaging would discard as noise. A further structural result—the *encapsulation principle*—establishes that the two strategies are not alternatives but hierarchical levels of a single architecture: every internal model is a compressed encoding of statistical regularities accumulated through an LLN-process at a longer timescale, so that the law of large numbers is never abandoned but descends to a lower floor of the system’s organization. The paper introduces the *survivorship principle*—that persistence is a precondition of observability rather than a goal of systems—and demonstrates that the apparent “drive toward survival” in complex systems is a selection artifact rather than a causal property. The framework unifies results from probability theory, non-equilibrium thermodynamics, predictive processing, and evolutionary biology within a single formal architecture, generating testable predictions about the relationship between agent cost, representational selectivity, and persistence strategy across domains.

Keywords: law of large numbers, complex systems, persistence, replaceability, Kolmogorov complexity, structural viability, predictive processing, survivorship bias, cognitive heuristics, non-equilibrium thermodynamics, agent cost

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1 Introduction

The law of large numbers (LLN) is among the most consequential results in probability theory. In its simplest formulation, it states that the arithmetic mean of a sequence of independent, identically distributed random variables converges to the expected value as the number of observations grows without bound [1–3]. This result underpins statistical inference, actuarial science, Monte Carlo methods [4], and the philosophical interpretation of probability as long-run relative frequency.

The mathematical conditions under which the LLN holds are well understood: independence (or sufficiently weak dependence), identical distribution (or bounded variance divergence), and the existence of a finite expected value [5–8]. When these conditions are met, convergence is assured. The question addressed in this paper is what happens when they are *systematically* violated—not by pathological mathematical construction, but by the structural requirements of systems that persist in changing environments.

The central argument proceeds in four stages. First, we establish that the conditions required by the LLN are incompatible with the structural requirements of complex adaptive systems, and show that intermediate convergence theorems (ergodic, martingale, mixing) do not resolve the problem for systems with high agent cost. Second, we introduce a formal measure of *agent replaceability*—the degree to which a system component’s structural contribution can be recovered after loss—and show that decreasing replaceability generates increasing pressure toward internal modeling as a substitute for external statistical averaging. Third, we develop a complementary criterion based on Kolmogorov complexity: the transition from statistical to model-based strategies occurs when the environment contains compressible structure at the agent’s operational scale that averaging would destroy. Fourth, we demonstrate that the cognitive and behavioral phenomena conventionally classified as “biases” or “heuristics” [9, 10] are structurally optimal responses to operating in regimes where the LLN cannot be relied upon—optimal not in the circular sense that they promote persistence, but in the formally characterizable sense that they exploit compressible environmental regularities that the LLN-strategy discards. A further structural result—the *encapsulation principle*—establishes that the two strategies are not alternatives but hierarchical levels: every internal model is a compressed encoding of statistical regularities accumulated through an LLN-process at a longer timescale, so that the LLN never disappears but recedes to a lower floor of the architecture. The core structure is: *many cheap trials at the training timescale* $\rightarrow$ *compression into model parameters* $\rightarrow$ *one costly action at the deployment timescale*—a pipeline that repeats at every level from evolution to individual motor acts (Figure 3).

A pivotal observation motivates the entire framework: we do not observe systems that fail to persist. This *survivorship principle* entails that persistence is not a property systems “strive for” but rather a precondition for any system’s being available for observation or analysis. The apparent teleology—the impression that systems “want” to survive—is a projection by the observer, an instance of the very cognitive biases that the framework explains [11, 12].

The paper connects literatures that have developed largely independently: probability theory and the formal conditions of the LLN [2, 3]; non-equilibrium thermodynamics and the costs of structural persistence [13–15]; predictive processing and the computational architecture of bounded cognition [16–20]; and error management theory and the evolutionary optimality of systematic bias [11, 12, 21]. The unifying concept is that the transition from statistical to model-based strategies is not a contingent feature of particular evolutionary lineages but a structural necessity derivable from the formal properties of persistence under irreplaceability.

2 The Law of Large Numbers: Formal Statement and Conditions

We begin with precise statements of the LLN to identify exactly which conditions complex systems violate.

2.1 Weak and Strong Forms

Definition 2.1 (Sample Mean). Let $X_1, X_2, \dots$ be a sequence of random variables. The *sample mean* after n observations is

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i. \quad (1)$$

The sample mean represents the running average of accumulated observations—the quantity that, under appropriate conditions, converges to the population parameter.

Theorem 2.2 (Weak Law of Large Numbers [6]). *Let $X_1, X_2, \dots$ be independent and identically distributed (i.i.d.) random variables with finite expected value $\mathbb{E}[X_i] = \mu$. Then for every $\varepsilon > 0$,*

$$\lim_{n \rightarrow \infty} \mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) = 0. \quad (2)$$

This states that the probability of the sample mean deviating from the true mean by any fixed amount, however small, can be made arbitrarily close to zero by taking sufficiently many observations. It does not guarantee that any particular finite sequence of observations will have converged—only that the probability of non-convergence vanishes in the limit.

Theorem 2.3 (Strong Law of Large Numbers [5]). *Under the same conditions as Theorem 2.2,*

$$\mathbb{P}\left(\lim_{n \rightarrow \infty} \bar{X}_n = \mu\right) = 1. \quad (3)$$

The strong law makes a sharper claim: with probability one, the sample mean converges to the true mean. Deviations may occur, but they occur only finitely many times for any fixed tolerance.

2.2 Necessary Conditions

The LLN requires three conditions, each of which has precise mathematical content:

- (i) **Independence** (or sufficiently weak dependence). Each observation must be uninfluenced by previous observations. Extensions to weakly dependent sequences exist [22], but some form of asymptotic independence remains necessary.
- (ii) **Identical distribution** (or controlled heterogeneity). All observations must be drawn from the same probability distribution. If the underlying distribution changes over time, the target of convergence itself shifts, potentially faster than the average can track.
- (iii) **Existence of finite expectation**. The expected value $\mathbb{E}[X_i] = \int x dF(x)$ must be finite. Distributions with heavy tails, such as the Cauchy distribution or Pareto distributions with shape parameter $\alpha \leq 1$, violate this condition [2, 23].

3 Structural Persistence and the Violation of LLN Conditions

3.1 Persistence as Structural Invariance

Following [24], we define persistence formally:

Definition 3.1 (Persistence). A system S *persists* over interval $[t_0, t_0 + \Delta t]$ if its structure Ω_S remains within a specified invariance class $\mathcal{I}_S$ throughout the interval:

$$\Omega_S(t) \in \mathcal{I}_S \quad \text{for all } t \in [t_0, t_0 + \Delta t]. \quad (4)$$

Here Ω_S denotes the equivalence class of microstates under a physically meaningful coarse-graining operation, and $\mathcal{I}_S$ is the set of structural configurations compatible with the system's continued identity.

A cell persists so long as its metabolic, membrane, and genetic organization remain within the bounds compatible with cellular function. The invariance class $\mathcal{I}_S$ captures the range of structural variation consistent with continued functioning—it need not be a single fixed state.

3.2 Thermodynamic Necessity of Transformation

Any system maintaining organization against thermodynamic equilibration must export entropy, and this necessarily transforms environmental structure [24]:

Proposition 3.2 (Transformational Necessity [24]). *Let S be an open system maintaining structure $\Omega_S \in \mathcal{I}_S$ against thermodynamic equilibration. Then there exists some subsystem E of the environment such that E undergoes structural transformation at a rate $T_E > 0$.*

The transformation rate scales with structural complexity. For trophic systems, if η denotes the efficiency of energy transfer between trophic levels ($\eta \approx 0.1$; see [25]), then a system at trophic level n requires transformation at rate

$$T_n = \frac{T_0}{\eta^n}, \quad (5)$$

where T_0 is the primary production rate. A tertiary consumer requires approximately 1000 times the structural transformation of primary producers, establishing a direct link between structural complexity and agent cost.

3.3 Why Persistence Violates LLN Conditions

3.3.1 Violation of Independence

A persisting system's state at time $t + 1$ is constrained by its state at time t . Observations are filtered through the system's current structural configuration, which was itself shaped by prior observations. If P_t denotes the perceived configuration at time t , then

$$P_{t+1} = f(P_t, A, C, M) \quad (6)$$

as established in [26], where A denotes the attention function, C the cultural-cognitive filter, and M the moral-evaluative constraint. The sequence $\{P_t\}$ is a history-dependent process, not a sequence of independent samples.

3.3.2 Violation of Identical Distribution

The environment of a persisting system is non-stationary. Seasons change, competitors adapt, resources deplete, and the system's own actions alter the environment. The random variables $X_1, X_2, \dots$ are not identically distributed; the distribution from which observations are drawn shifts over time, and is partially constituted by the agent's own behavior.

3.3.3 Violation of Finite Expectation

Many quantities relevant to persistence exhibit heavy-tailed distributions. Earthquake magnitudes, extinction events, financial crashes, and predator encounter rates follow distributions with tails heavier than exponential [23, 27–30]. A single extreme event can destroy the system regardless of favorable averages. This is precisely the regime where the LLN provides no guarantees: convergence of the sample mean is irrelevant if the population mean does not capture the risk of ruin.

3.4 Why Intermediate Convergence Theorems Do Not Resolve the Problem

It would be natural to object that the violations cataloged above are well-known, and that extensions of the LLN to dependent, non-stationary, and heavy-tailed settings already exist. Ergodic theorems [31] guarantee convergence for stationary processes without requiring independence. Martingale convergence theorems [32] provide convergence guarantees for dependent sequences satisfying a conditional expectation constraint. Mixing conditions [33] quantify the degree of asymptotic independence required for convergence under dependence.

These extensions are mathematically important, but they do not resolve the problem for complex adaptive systems, for three reasons. First, the ergodic theorem requires stationarity—precisely the condition violated by system–environment coupling (Section 3.3.2). A system whose actions alter its environment faces a non-stationary process for which ergodic convergence does not apply. Second, mixing conditions require that correlations decay sufficiently rapidly with temporal separation; but the path dependence induced by internal model maintenance (Section 3.3.1) produces correlations that may persist indefinitely—each observation is filtered through a model built from all preceding observations. Third, and most fundamentally, even when weaker convergence theorems do apply, the *rate* of convergence may be too slow for systems with high agent cost. A system that will converge to accurate estimates in 10^6 generations is not helped by this guarantee if it must act correctly within a single lifetime to persist. The issue is not that convergence is mathematically impossible but that it is *temporally inaccessible* for systems whose agents cannot be replaced fast enough to wait for it.

Table 1: Systematic violations of LLN conditions in complex adaptive systems.

LLN Condition	Mechanism of Violation	Structural Source
Independence	Path-dependent observation	Internal model maintenance
Identical distribution	Non-stationary environment	System–environment coupling
Finite expectation	Heavy-tailed risk	Catastrophic event vulnerability

The violations form a mutually reinforcing triad: path dependence (maintaining an internal model) is a response to non-stationarity (shifting environment), which is exacerbated by heavy-tailed risks that make waiting for convergence fatally dangerous.

4 Agent Replaceability and Agent Cost

The central claim of this paper is that the transition from statistical to model-based persistence strategies is governed by a single structural parameter: the *replaceability* of the individual agent within the system.

4.1 Formal Definitions

Definition 4.1 (Agent Replaceability). Let S be a system composed of agents $\{a_1, \dots, a_N\}$, and let $\mathcal{I}_S$ denote the system’s viability set. The *replaceability* of agent a is

$$\mathcal{R}(a) = \frac{|\{a' \in \mathcal{A} : \mathcal{I}_S(S_{a \rightarrow a'}) \cap \mathcal{I}_S(S) \neq \emptyset\}|}{|\mathcal{A}|}, \quad (7)$$

where $\mathcal{A}$ is the pool of available replacement agents, and $S_{a \rightarrow a'}$ denotes the system with agent a replaced by a' .

Replaceability thus measures the proportion of available agents that could substitute for a without removing the system from its viability set. The pool $\mathcal{A}$ is defined as the set of agents producible by the system’s generative mechanisms within one characteristic persistence timescale—for a biological system, this means organisms that the reproductive and developmental processes of the population can produce within one generation. This specification is necessary because replaceability is a property of the system’s regenerative capacity, not an abstract combinatorial quantity. In a bacterial population, $\mathcal{A}$ consists of all bacteria the colony can produce by division within one generation (~ 20 minutes); since each performs essentially the same metabolic functions, $\mathcal{R}(a) \approx 1$. In a mammalian nervous system, each neuron occupies a specific position shaped by decades of experience-dependent plasticity, placing $\mathcal{R}(a)$ substantially below 1.

Definition 4.2 (Agent Cost). The *cost* of agent a to system S is

$$\mathcal{C}(a) = (1 - \mathcal{R}(a)) \cdot \tau_{\text{recovery}}(a), \quad (8)$$

where $\tau_{\text{recovery}}(a)$ is the time required for the system to restore the functional capacity lost when a is removed, measured in units of the system’s characteristic persistence timescale.

Agent cost integrates two components: how difficult it is to find a substitute ($1 - \mathcal{R}(a)$) and how long the system must operate in a degraded state (τ_{recovery}). A bacterium dividing every 20 minutes has $\mathcal{C}(a) \approx 0$. A human whose neural architecture required decades to form has $\mathcal{C}(a) \gg 1$.

A boundary case deserves explicit discussion. For cortical neurons in adult mammals, $\mathcal{A}$ is effectively empty: adult neurogenesis is confined to a few regions (dentate gyrus, olfactory bulb), and most cortical neurons cannot be replaced at all. This yields $\mathcal{R}(a) = 0$ trivially, which is formally correct but uninformative. The resolution is that the system’s response to irreplaceable agents is not computed through the replaceability formula but through the *architectural consequences* of $\mathcal{R}(a) = 0$: the brain invests massively in redundancy (distributed representations, graceful degradation), protective infrastructure (skull, blood-brain barrier, glia), and compensatory plasticity (functional reorganization after damage). These are precisely the strategies predicted by the framework for the limit $\mathcal{C}(a) \rightarrow \infty$: when replacement is impossible, the system must prevent loss entirely or redistribute function after loss. The “degenerate” case is not a failure of the definition but its most extreme prediction.

4.2 Replaceability and Structural Connectivity

Replaceability is relational, not intrinsic. An agent participating in many cooperative interactions—contributing to joint states unavailable to other agents alone—is harder to replace. This connects to the Law of Cooperation [34]: self-organizing systems require synergy. The degree of synergistic participation provides a motivated upper bound on replaceability (not a theorem, but an inequality whose validity follows from the observation that each synergistic interaction in which a participates constitutes a joint state that a replacement agent must reproduce):

$$\mathcal{R}(a) \leq 1 - \frac{\text{synergistic interactions involving } a}{\text{total synergistic interactions in } S}. \quad (9)$$

This bound becomes tight in the idealized case where each synergistic interaction depends on exactly one irreplaceable participant; in general, redundancy among synergistic partners loosens it.

5 Scale-Dependent Compressibility and the Information-Theoretic Criterion

The preceding sections established that the LLN’s conditions are violated by persisting systems, and that agent cost determines which persistence strategy is viable. This section introduces a complementary criterion—grounded in Kolmogorov complexity—that provides a formal characterization of *when* the transition between strategies occurs, resolving the indeterminacy of the critical threshold $\mathcal{C}^*$ and addressing the question of what, precisely, internal models exploit that statistical averaging cannot.

5.1 Kolmogorov Complexity and Scale

The Kolmogorov complexity $K(x)$ of a finite string x is the length of the shortest program that produces x on a universal Turing machine [35, 36]. A string is *incompressible* (algorithmically random) if $K(x) \approx |x|$: no program shorter than the string itself can generate it. A string is *compressible* if $K(x) \ll |x|$: it contains regularities that a program can exploit to produce it from a shorter description.

A fundamental but under-appreciated property of Kolmogorov complexity is that compressibility is *relative to the description language*—and therefore relative to the scale of observation. Consider the molecular dynamics of a gas. At the scale of individual particle trajectories, the microstate sequence is effectively incompressible: specifying the position and momentum of 10^{23} particles at each timestep requires a description approximately as long as the data itself. At the macroscopic scale, the same physical process is captured by a handful of compressible quantities—temperature, pressure, volume—related by the ideal gas law. The data that was noise (incompressible) at the micro-scale has become information (compressible) at the macro-scale. Coarse-graining is, in this precise sense, a change of description language that transforms incompressible sequences into compressible ones.

5.2 The LLN as a Compressibility Theorem

This perspective reveals the LLN’s deep informational content. The sample mean $\bar{X}_n$ is a *compression* of the data $(X_1, \dots, X_n)$: it reduces n observations to a single number. The LLN guarantees that this compression is asymptotically *sufficient*—that $\bar{X}_n$ captures all the recoverable information in the sequence, and the residuals $(X_i - \bar{X}_n)$ are incompressible noise. In information-theoretic terms, the LLN identifies the unique partition of the data into compressible signal (the mean) and incompressible noise (the deviations), and guarantees that this partition is optimal as $n \rightarrow \infty$.

When the i.i.d. conditions hold, the environment’s compressible content at the population scale is fully captured by the population parameters $(\mu, \sigma^2, \text{etc.})$, and the LLN-strategy—which extracts these parameters through averaging—is informationally complete. No internal model can do better, because there is no additional compressible structure to exploit.

5.3 Structure That Averaging Destroys

The critical transition occurs when the environment contains compressible structure at the agent’s operational scale that the LLN-strategy would treat as noise. This happens in precisely those regimes where the LLN conditions are violated:

- (i) **Dependence implies temporal compressibility.** If observations are dependent, then the sequence $(X_1, X_2, \dots)$ contains temporal regularities—patterns in the ordering of observations—that a model can exploit for prediction. Averaging destroys temporal structure: $\bar{X}_n$ is invariant under permutation of the X_i , so any information encoded in the sequence’s ordering is discarded. For a system facing a non-i.i.d. environment, the temporal structure may carry persistence-relevant information (predator approach patterns, seasonal resource fluctuations, social dynamics) that averaging obliterates.
- (ii) **Non-stationarity implies contextual compressibility.** If the distribution shifts over time, then the environment contains compressible structure in the form of regime descriptions: “in context A , the distribution is F_A ; in context B , F_B .” A model that represents these contexts separately compresses the data more efficiently than a single grand average, which conflates distinct regimes. The difference between the model-strategy and the LLN-strategy is precisely the difference between context-sensitive compression and context-blind averaging.
- (iii) **Heavy tails imply extremal compressibility.** In heavy-tailed distributions, the tail events carry disproportionate information about system-relevant quantities (survival probability, catastrophic risk). A model that represents the tail structure—that distinguishes between “typical fluctuation” and “existential threat”—compresses the relevant information more efficiently than the sample mean, which is dominated by the bulk of the distribution and is insensitive to tail structure.

5.4 The Scale-Compressibility Criterion

These observations yield a formal criterion for the strategy transition that complements the agent-cost argument of Section 4:

Definition 5.1 (Persistence-Relevant Compressibility). Let S be a system operating at scale s , and let $\mathbf{X}_s = (X_1, X_2, \dots)$ denote the sequence of environmental observations at scale s . The *persistence-relevant compressibility* at scale s is

$$\kappa_s = \frac{|\mathbf{X}_s| - K_s(\mathbf{X}_s)}{|\mathbf{X}_s|}, \quad (10)$$

where $K_s(\mathbf{X}_s)$ is the Kolmogorov complexity of $\mathbf{X}_s$ evaluated relative to a description language appropriate to scale s , restricted to regularities relevant to the system’s persistence (i.e., regularities in the viability-relevant components of the environment).

A fundamental objection arises immediately: Kolmogorov complexity is uncomputable (a consequence of the halting problem), and the triply conditioned quantity $K_s(\mathbf{X}_s)$ —restricted by scale, description language, and persistence relevance—is *a fortiori* uncomputable. The definition therefore cannot serve as an operational criterion in its exact form.

We address this objection at two levels. First, the role of κ_s in the framework is *structural*, not computational: it identifies *what* the model-strategy exploits (compressible environmental regularities at the agent’s operational scale) and *why* the LLN-strategy cannot substitute (because averaging destroys scale-specific structure). This structural identification does not require computing κ_s exactly, any more than the role of entropy in thermodynamics requires computing the exact microstate count of a gas. Second, computable approximations exist and are well-studied. The Minimum Description Length (MDL) principle [37] provides a computable upper bound on Kolmogorov complexity by restricting the class of programs to a specified model family. Normalized compression distance [36] provides an empirically tractable approximation. In practice, the question “does this environment contain exploitable structure beyond what averaging captures?” is answered not by computing K_s but by comparing the predictive performance of

a model-based system against a pure averaging system—which is precisely what the robotics experiments of Section 8 demonstrate.

When $\kappa_s \approx 0$, the environment at scale s is effectively incompressible for persistence purposes: there is no exploitable structure beyond what statistical averaging can extract. The LLN-strategy suffices. When $\kappa_s > 0$, the environment contains compressible persistence-relevant structure at scale s that a model can exploit and that averaging would destroy. The model-strategy becomes advantageous—and, when agent cost is sufficiently high, necessary.

Proposition 5.2 (Scale-Compressibility Transition). *The LLN-strategy is sufficient for system S at scale s if and only if the persistence-relevant compressibility κ_s at scale s is negligible relative to the information extractable by averaging. When κ_s exceeds a threshold determined by agent cost $\mathcal{C}$, the model-strategy is required: the system must construct an internal representation that exploits the compressible structure rather than averaging over it.*

This proposition provides the formal content that the critical threshold $\mathcal{C}^*$ in Proposition 6.3 lacked: the transition occurs not at a fixed agent cost but at the point where agent cost is high enough that the system cannot afford to discard the compressible structure that averaging would destroy.

Relationship between the two criteria. The framework now contains two transition criteria: the *agent-cost criterion* ($\mathcal{C} > \mathcal{C}^*$, Section 4) and the *compressibility criterion* (κ_s exceeds a threshold). These are not independent conditions but complementary aspects of a single transition. The agent-cost criterion determines *whether* the system must model; the compressibility criterion determines *whether* modeling is possible. A high-cost agent in a genuinely random environment ($\mathcal{C}$ large, $\kappa_s \approx 0$) faces the worst case: the LLN-strategy is too slow and the model-strategy has nothing to exploit. Such systems are predicted to be evolutionarily unstable—they cannot persist at either pole. In practice, this combination is rare precisely because high-cost agents tend to evolve in environments with exploitable structure (the environments that selected for high cost in the first place). When the combination does arise—a specialist organism facing a sudden, structurally random catastrophe—the predicted outcome is extinction, which is indeed what the fossil record shows for specialist taxa encountering bolide impacts or massive volcanism.

5.5 Resolving the Optimality Circularity

The scale-compressibility framework also resolves a potential circularity in the claim that cognitive biases are “optimal.” The circularity arises if optimality is defined solely by persistence: biases are optimal because biased organisms persisted, and we know they are optimal because we observe them persisting. The compressibility criterion provides a non-circular exit. Biases are optimal not because they promote persistence (which would be circular) but because they *exploit compressible structure at the agent’s operational scale*—structure that the LLN-strategy would treat as noise and average away. This is a formally characterizable, independently verifiable property: one can, in principle, measure the Kolmogorov complexity of the environmental sequence at the agent’s operational scale and determine whether the agent’s representational distortions track the compressible regularities rather than the incompressible noise.

The confirmation bias, for instance, is not merely “a bias that happened to promote persistence”; it is a compression heuristic that exploits the temporal autocorrelation of environmental states (the fact that environments tend to remain similar over short timescales). This autocorrelation is a compressible regularity at the agent’s operational timescale. The bias exploits it; the LLN-strategy, which treats successive observations as exchangeable, cannot.

6 Two Persistence Strategies

6.1 The LLN-Strategy: Population-Level Statistical Averaging

Definition 6.1 (LLN-Strategy). A system employs the *LLN-strategy* when persistence is achieved through the statistical averaging of outcomes across a large population of individually expendable agents.

The paradigmatic example is a bacterial population. Individual bacteria encounter variable environments, and many perish. But the population as a whole exhibits stable properties because the LLN ensures that population averages converge. The LLN-strategy is available when: (a) $\mathcal{R}(a) \approx 1$, (b) $N \gg 1$, and (c) generation time is short relative to environmental change.

6.2 The Model-Strategy: Individual-Level Internal Simulation

Definition 6.2 (Model-Strategy). A system employs the *model-strategy* when persistence is achieved through internal representations that simulate environmental dynamics, enabling prediction and preemptive action by individual agents.

A clarification is essential here. The model-strategy does not represent an abandonment of probabilistic reasoning; it represents a transition in the *source of statistical power*. Under the LLN-strategy, statistical power derives from sample size: many observations converge to accurate estimates. Under the model-strategy, statistical power derives from model structure: accumulated priors—evolutionary, developmental, cultural—enable informative inference from small samples. This is precisely the distinction between frequentist and Bayesian approaches to uncertainty. A bacterial population is, in effect, a Monte Carlo method: it generates millions of independent samples and relies on the LLN to ensure that population-level averages converge. A mammalian brain is, in effect, a Bayesian neural network with trained priors: it generates predictions from a structured model and updates them with whatever data arrives, however sparse [16, 17, 20].

The Bayesian character of the model-strategy is empirically well-supported. Multisensory integration experiments demonstrate that the brain combines visual and tactile signals with weights inversely proportional to each channel’s noise—precisely as Bayesian inference prescribes [38]. Dopaminergic neurons encode reward prediction errors that correspond to Bayesian updating signals [16]. Hippocampal replay during sleep generates sequences whose frequencies correlate with their expected utility, consistent with sampling from a learned posterior distribution [18].

The critical insight is that Bayesian inference with strong priors is informationally equivalent to LLN convergence with large samples—but achieves the same effective certainty from far fewer observations, at the cost of prior dependence. This cost is the structural distortion identified in [26]: the priors that enable rapid inference from small samples are themselves a compression of environmental structure, and this compression necessarily introduces systematic divergence between the perceived and actual environment.

Internal modeling is also thermodynamically expensive. By Landauer’s principle [39, 40], the minimum energy cost of maintaining k bits of predictive model is

$$E_{\min} = k \cdot k_B T \ln 2, \quad (11)$$

where T is the environmental temperature and k_B is Boltzmann’s constant. Biological neural systems operate at approximately 10^4 times this minimum per synaptic modification [41]—a substantial metabolic investment that only systems with high agent cost can justify.

The selectivity of representational compression is quantified as

$$\sigma = 1 - \frac{H(P)}{H(R)}, \quad (12)$$

where $H(P)$ is the entropy of the perceived configuration and $H(R)$ is the entropy of the reference environment [42, 43]. As σ increases, the representation becomes more compressed—the priors become stronger—preserving variance in domains critical for persistence while suppressing variance in domains judged irrelevant [26].

6.3 The Fundamental Trade-Off

Proposition 6.3 (Strategy Transition). *Let S be a system with characteristic agent cost $\mathcal{C}$. There exists a critical threshold $\mathcal{C}^*$ such that:*

- (i) *If $\mathcal{C} < \mathcal{C}^*$, the LLN-strategy is sufficient for persistence.*
- (ii) *If $\mathcal{C} > \mathcal{C}^*$, the system must develop internal models with selectivity $\sigma > 0$.*

The following argument establishes the plausibility of this claim and characterizes the mechanism of transition; it is a heuristic derivation rather than a rigorous proof, as a complete proof would require a fully specified dynamical model of system–environment interaction.

Argument. Suppose $\mathcal{C}(a) > 0$ for some agent a . The system’s viability set $\mathcal{I}_S$ is compromised during the interval $[t_{\text{loss}}, t_{\text{loss}} + \tau_{\text{recovery}}]$ following the loss of a . For the LLN-strategy, the expected number of agent losses per unit time is $N \cdot p_{\text{loss}}$. Under the simplifying assumption that agent losses occur as a Poisson process and that each loss independently reduces viability by a factor proportional to $\mathcal{C}_{\text{avg}}$, the probability that the system remains viable satisfies

$$\mathbb{P}(\text{viable at time } t) \leq \exp(-N \cdot p_{\text{loss}} \cdot \mathcal{C}_{\text{avg}} \cdot t). \quad (13)$$

When $\mathcal{C}_{\text{avg}} \approx 0$, this probability remains close to 1. When $\mathcal{C}_{\text{avg}}$ is large, it decays rapidly.

To reduce the effective loss rate, the system must either reduce p_{loss} for high-cost agents or increase $\mathcal{R}(a)$ through redundancy. The former requires that high-cost agents predict and avoid threats rather than encountering them stochastically—this is precisely the model-strategy. The latter requires maintaining redundant copies, itself requiring additional structural transformation proportional to $\mathcal{C}(a)$ by Proposition 3.2.

The critical threshold $\mathcal{C}^*$ is reached when the expected cost of agent loss under the LLN-strategy exceeds the system’s capacity to maintain viability through replacement alone. A precise characterization of $\mathcal{C}^*$ as a function of system parameters (N , p_{loss} , environmental volatility) remains an open problem; Section 5 provides a complementary characterization in terms of scale-dependent compressibility that partially addresses this gap. $\diamond$

7 Chemotaxis: The Transitional Case

Bacterial chemotaxis provides a uniquely informative example because it exhibits features of both persistence strategies simultaneously.

7.1 The Run-and-Tumble Mechanism

Escherichia coli navigates chemical gradients through a biased random walk [44–46]. Recent reviews have further emphasized the remarkable sophistication of bacterial chemotactic signaling and its clinical significance [47]. The bacterium alternates between *runs* (straight-line swimming, ~ 1 second) and *tumbles* (random reorientation, ~ 0.1 seconds). In a chemical gradient, tumble frequency decreases when swimming up-gradient and increases when swimming down-gradient [48, 49], producing net movement toward the attractant.

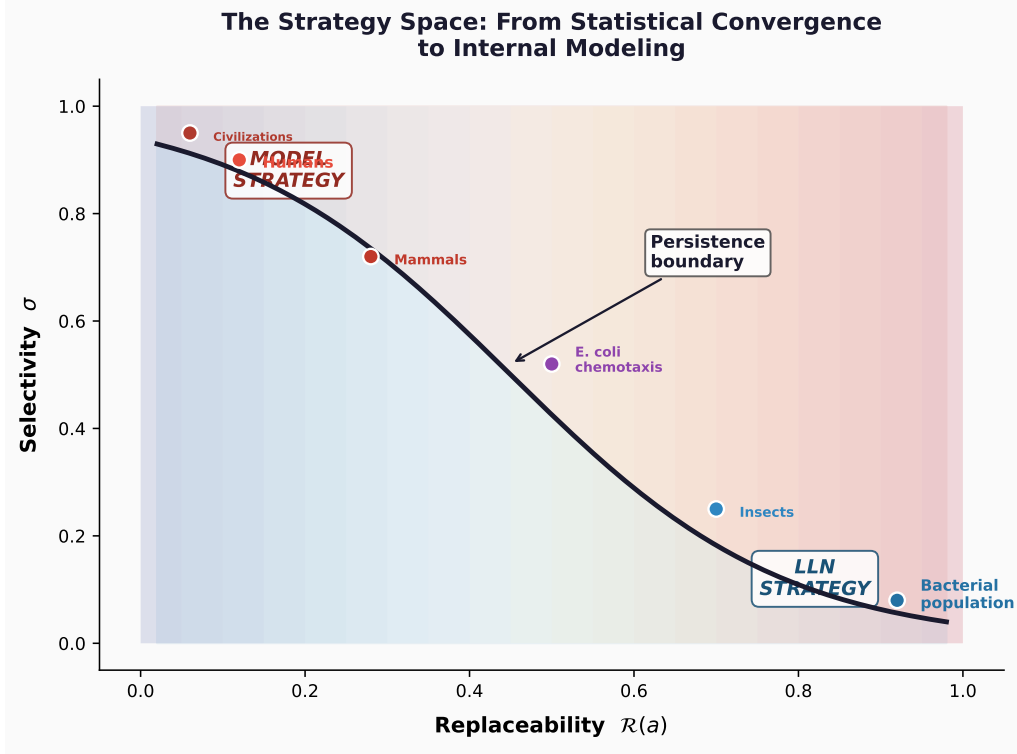


Figure 1: **The strategy space.** As replaceability $\mathcal{R}(a)$ decreases (left), required selectivity σ increases (up), reflecting the transition from LLN-strategy (lower right) to model-strategy (upper left). Exemplar systems are positioned according to their characteristic replaceability and selectivity values.

7.2 Dual Strategy in a Single System

At the *population level*, chemotaxis operates as an LLN-strategy: the population migrates toward the attractant with a drift velocity reflecting the statistical average of many biased random walks. At the *individual level*, each bacterium performs a temporal comparison: it compares current attractant concentration with that sensed approximately one second earlier [49]. This is a primitive internal model—a single-bit memory recording “better” or “worse.”

Remark 7.1 (The Minimal Model). The chemotactic temporal comparison is perhaps the simplest possible internal model. It does not represent spatial structure or predict future states; it merely registers the direction of recent change. Yet this minimal model suffices to convert an unbiased random walk into a biased one, producing directed motion at the population level.

The agent cost of an individual bacterium is low ($\mathcal{C}(a) \approx 0$), so the population relies primarily on the LLN-strategy. But the slight advantage conferred by temporal comparison—a marginal increase in individual survival probability—is sufficient for selection to maintain the modeling capacity.

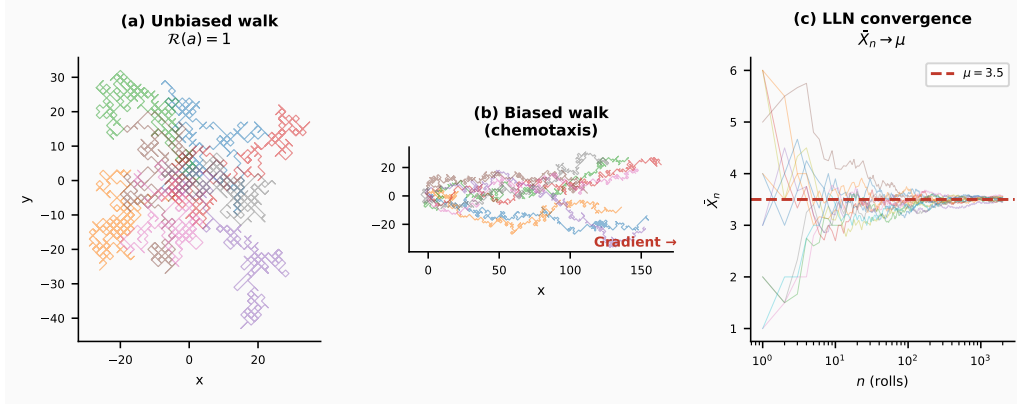


Figure 2: **From unbiased random walk to LLN convergence.** (a) Unbiased walks: no net displacement ($\mathcal{R}(a) = 1$). (b) Biased walks (chemotaxis): slight bias produces directed population-level motion. (c) Classical LLN convergence: sample means converge to $\mu = 3.5$.

8 Motor Control as Model-Strategy: The Cat Jump

Where chemotaxis illustrates the minimal threshold of internal modeling, the ballistic motor control of a predatory mammal illustrates the model-strategy operating at full capacity. The case of the cat jump is instructive because it makes the Bayesian architecture of the model-strategy concrete and falsifies any impression that “internal modeling” is a metaphorical rather than a literal description.

8.1 The Computational Problem

A cat leaping to a target must estimate distance (via binocular disparity and motion parallax), compute the required force and angle of launch, coordinate dozens of muscles across four limbs with millisecond-level timing, and commit to the entire motor plan before leaving the surface—since the jump lasts 300–500 milliseconds and visual feedback latency is approximately 100 milliseconds, in-flight correction is largely impossible [38]. Robotics laboratories have invested decades in approximating this capability and remain far from the performance of an ordinary house cat.

8.2 Bayesian Architecture in the Cerebellum

The cat does not solve the differential equations of ballistic motion. Instead, the cerebellum maintains two classes of internal model [38]: *forward models* that predict the sensory consequences of motor commands (“if I push off with this force, I will land approximately there”), and *inverse models* that compute the motor commands required for a desired outcome. These models are not derived analytically; they are trained through thousands of developmental trials. Each trial generates a prediction error—the difference between predicted and actual sensory outcome—transmitted via climbing fibers from the inferior olive, which modifies Purkinje cell synapses through a process formally analogous to gradient descent.

The visual system does not compute a trajectory; it classifies the jump into a region of parameter space (distance, height, surface texture) that retrieves the corresponding motor program via the cerebellar model. The characteristic head-bobbing of a cat before a jump is not hesitation but active information gathering—motion parallax extraction to reduce the uncertainty of the distance estimate, consistent with the behavior of a Bayesian agent seeking to reduce posterior variance before committing to an irreversible action (though the Bayesian optimality of the specific bobbing pattern has not been formally demonstrated).

8.3 Compression of Control Space

A crucial feature of the architecture is the radical compression of the control space. The spinal cord’s central pattern generators produce the detailed temporal sequence of muscle activations; the brain specifies only a low-dimensional set of parameters—force, direction, timing—that the spinal circuits elaborate into the full motor program. This is compression in precisely the sense formalized in Section 5: the high-dimensional motor command sequence has high Kolmogorov complexity when described muscle by muscle, but low complexity when described in the compressed parameter space that the cerebellar model operates in. The model-strategy works because the physics of ballistic jumps contains compressible structure—the relationship between launch parameters and landing position is lawful, not random—and the cerebellar model has learned to exploit this compressibility.

8.4 Structural Distortion in Action

The cat sometimes misses. The internal model is imperfect—it is a compression, and compression entails loss. But the cat misses far less often than would be expected from a system that has never solved a single equation of motion. This is the power of the model-strategy expressed in its purest form: an approximately correct answer immediately, rather than an exact answer that would require computational resources no biological system possesses. The structural distortion ($\sigma > 0$) is not a failure but a design constraint: the model preserves the variables critical for survival (distance, surface stability, escape routes) while discarding variables irrelevant to persistence (air resistance coefficients, precise gravitational acceleration). The distortion is patterned by the agent’s persistence requirements—precisely as the Structural Distortion Principle predicts [26].

8.5 Embodied Computation and the Correct Representation Space

A deeper point emerges from the cat example. The cat’s computational system is not confined to the brain: the elasticity of tendons, the inertia of limbs, and the gyroscopic stabilization provided by the tail all participate in the “computation” without being symbolically represented. This is what cognitive science terms *embodied computation*—the body is part of the model, not merely its output device. The relevant representation space is not Cartesian coordinates (positions, forces, angles) but the space of *motor synergies*: low-dimensional coordination patterns for which no adequate mathematical formalism yet exists. The brain operates in this compressed space, and the spinal cord decompresses into the high-dimensional muscle activation space. The environment at the agent’s operational scale is compressible precisely because physics is lawful, and the model exploits this compressibility in its native coordinate system rather than in ours.

This matters because it demonstrates that the model-strategy is not merely “symbolic reasoning under time pressure.” It is a fundamentally different computational architecture: continuous rather than discrete, parallel rather than sequential, and natively stochastic. Uncertainty is not added to a deterministic model as an afterthought; it is the medium in which computation occurs. Neural noise—which would be a defect in a symbolic computer—is a computational resource in a sampling-based Bayesian architecture.

8.6 Independent Confirmation from Robotics

Recent advances in neural-network-based robot locomotion provide a striking independent test of the framework’s predictions. Classical robot control uses the LLN-strategy’s computational analogue: explicit dynamical equations, analytical trajectory planning, and symbolic optimization. The results are slow, fragile, and confined to flat surfaces. Neural-network controllers trained by reinforcement learning—the model-strategy implemented on silicon—produce dramatically superior locomotion [50, 51].

The most significant result for the present framework comes from quadrupedal robots trained with “viability” (persistence) as their sole objective function [52]. These robots spontaneously invented gait transitions—switching between walking, trotting, and pronking (a gazelle-like bounding gait) when encountering obstacles—without any instruction to do so. Of three possible training objectives (energy efficiency, load reduction, and viability), *only viability produced spontaneous gait transitions*. This is a direct experimental confirmation of the central thesis: the structure of the persistence problem itself—high cost of failure, unreliable terrain—generates the model-strategy and its characteristic properties, independently of the physical substrate.

Full-scale humanoid robots with transformer-based neural controllers now demonstrate outdoor walking without falls, resistance to external perturbations, and emergent behavioral patterns such as arm swinging—properties that no one programmed but that the model-strategy produces as a structural consequence of optimizing for persistence [51]. The trajectory of robotics is precisely the trajectory predicted by Figure 1: from the lower-right corner (analytical methods, LLN-strategy’s computational analogue) toward the upper-left (learned internal models, model-strategy).

8.7 Microgravity as Natural Experiment: When the Model’s Priors Become False

The cat’s cerebellar model encapsulates a fundamental prior: constant gravitational acceleration of 9.8 m/s^2 directed downward, with a stable “down” defined by the otolith organs of the vestibular system. Microgravity experiments provide a natural test of the framework’s predictions about model failure under distributional shift, and the results illuminate the hierarchical architecture of the model-strategy with remarkable precision.

In the 1950s, the U.S. Air Force Aerospace Medical Research Laboratory conducted a series of experiments in which cats were released during parabolic flights aboard Convair C-131 aircraft [53]. The results were striking: the righting reflex—the cat’s celebrated ability to land on its feet—was almost completely abolished in weightlessness. Without gravitational input to the otolith organs, the cerebellar forward model lost its primary reference signal, and the reflex that normally operates with near-perfect reliability became dysfunctional. This is the clearest possible demonstration of model failure from prior invalidation: the model is structurally intact, but its foundational input—the gravity vector—has been removed. In the terminology of the present framework, the distributional shift from $1g$ to $0g$ invalidates not the model’s parameters but its *architectural presupposition*—the existence of a preferred direction.

However—and this is the critical observation—the cats were not entirely helpless. Despite disorientation, they continued to attempt body rotations, suggesting that proprioceptive and motor components of the model retained partial functionality even when the vestibular component was invalidated [53]. The model did not fail as a monolithic unit; it degraded *hierarchically*, with the vestibular-dependent components failing while proprioceptive components continued to operate.

Most significantly, the formal experiments revealed not merely reflexive failure but the beginnings of higher-order adaptation. Gerathewohl and Stallings noted that cats continued to attempt body rotations despite disorientation, and the comprehensive review by Mori [54] established that across species, coordinated performance in microgravity can be compensated by visual function—mice, monkeys, and turtles all demonstrated visual compensation for vestibular loss. Informal documentation from a 2008 parabolic flight experiment at the Yuri Gagarin Cosmonaut Training Center corroborates this pattern: over ten consecutive parabolas (~ 230 seconds of cumulative weightlessness), a cat was observed to progressively adapt, learning to use objects in the cabin as graspable surfaces for reorientation.¹ This behavioral shift—from re-

¹The 2008 experiment (cat “Porculpa,” Il-76 parabolic flight from Star City, Moscow) is documented in informal video records rather than peer-reviewed literature. The observation is consistent with the formal findings of [53]

flexive flailing to instrumental use of environmental objects—is not cerebellar recalibration (the timescale is far too short for synaptic remodeling of Purkinje cells) but *cortical problem-solving*: the higher-order model recognized that the lower-order model was unreliable and generated an alternative strategy.

This behavioral sequence reveals the hierarchical nesting of the model-strategy operating in real time, not merely across evolutionary timescales:

- (i) **Level 1: Cerebellar model** (encapsulated evolutionary and developmental LLN). The righting reflex fails because its fundamental prior (gravity vector) is absent. This level cannot adapt on the timescale of the experiment—it requires extensive retraining through climbing fiber error signals.
- (ii) **Level 2: Cortical meta-model** (encapsulated lifetime experiential LLN). The cortex detects the discrepancy between the cerebellar model’s predictions and actual sensory feedback, reduces confidence in the Level 1 model, and generates an alternative strategy: using objects in the environment as anchors. This is Bayesian inference over model reliability—a hyperprior that governs which lower-level model to trust.
- (iii) **Level 3: Evolutionary architecture** (encapsulated deep-time LLN). The very capacity to switch from reflexive to exploratory behavior in novel environments is itself an evolved trait. Cats that could not transition from confident to cautious behavior in unfamiliar conditions were eliminated by natural selection. This architectural flexibility is the deepest layer of encapsulated statistics.

Comparative data from other species reinforce the pattern. A comprehensive review of animal disorientation in microgravity found that while fundamental vestibular reflexes are impaired in weightlessness, coordinated performance can be compensated by visual function in mice, monkeys, and turtles [54]. Fish exposed to sustained microgravity during space missions showed initial disorientation (tumbling), but sensory conflicts diminished significantly within 3–4 days, with cerebellar involvement in the recovery [54]. This 3–4 day timescale is consistent with cerebellar synaptic remodeling—the Level 1 model undergoing LLN-driven retraining in the new gravitational regime.

The microgravity data also generated a remarkable practical consequence that directly illustrates the cross-substrate transfer of model-strategy principles. In 1962, the same Air Force laboratory published a report on self-rotation techniques for astronauts in weightlessness, directly applying lessons learned from cats [55]. The first described technique was named the “Cat Reflex”—an explicit transfer of the biological model-strategy to human motor learning. Kane and Scher’s subsequent mathematical analysis modeled the cat’s body as two counter-rotating cylinders that conserve angular momentum [56]—providing the analytical framework that NASA adopted for astronaut training. The cat’s embodied computation, evolved over millions of years, was reverse-engineered into explicit models that humans could consciously learn.

Predictions for altered gravity. The framework generates specific, testable predictions for feline motor behavior under partial gravity (lunar or Martian stations with Earth-normal atmospheres):

- (a) **Lunar gravity ($1/6g$):** The gravitational vertical exists, so vestibular input is qualitatively correct but parametrically wrong. Initial errors should be *directional* (systematic overshoot due to excessive force calibrated for Earth gravity) rather than random, with magnitude proportional to $g_{\text{Earth}}/g_{\text{Moon}} \approx 6$. Cerebellar recalibration (Level 1 retraining) should restore near-normal performance within days to weeks—a parametric correction within an intact architecture.

and [54] but should be treated as anecdotal evidence.

- (b) **Martian gravity (0.38g):** Intermediate case. Smaller distributional shift predicts faster adaptation and smaller initial errors. This provides a quantitative test: the ratio of adaptation times (Moon/Mars) should approximate the ratio of parametric shifts.
- (c) **Microgravity (space station):** Qualitative architectural failure. Adaptation possible but limited by the fact that the cerebellar architecture itself is optimized for a gravitational environment. Long-term adaptation should plateau below Earth-gravity performance levels, with the deficit reflecting the cost of operating an architecture outside its design envelope—a direct test of Prediction P6.

9 Cognitive Biases as Structural Consequences of the Model-Strategy

When agent cost is high and the model-strategy is the only viable persistence mechanism, the internal model must operate under severe resource constraints, producing systematic patterns cataloged as “biases.”

9.1 The Structural Distortion Principle

The SDP [26] establishes that for any cognitive system with finite processing resources, there exists a trade-off between adaptive fitness and environment-tracking accuracy. The divergence between perceived and actual environmental structure can be measured by the Kullback-Leibler divergence:

$$D_{\text{KL}}(P\|R) = \sum_x P(x) \ln \frac{P(x)}{R(x)}, \quad (14)$$

which quantifies the information lost when the compressed representation P is used in place of R [43]. The SDP predicts that this divergence is *structured*: patterned by the agent’s cultural, moral, and volitional configuration, preserving detail in domains critical for persistence.

9.2 Error Management and the Optimality of Bias

Error management theory [11, 12] establishes that when costs of different error types are asymmetric, selection favors biases toward the less costly error. Within the present framework, this acquires a deeper interpretation: the asymmetry of error costs is itself a consequence of agent cost. When $\mathcal{C}(a)$ is high, the cost of a false negative (agent loss) is disproportionately large relative to false positives (wasted energy). The resource-rational analysis program [21] confirms that many apparent biases are optimal given computational constraints.

The confirmation bias [57] is a direct consequence of the perceptual configuration cycle [26]: information consistent with the existing configuration receives sustained attentional engagement; inconsistent information is rapidly categorized and dismissed. This is the system operating as designed by selection: maintaining a coherent internal model at minimal computational cost.

9.3 The “Law of Small Numbers” Reconsidered

Tversky and Kahneman [58] identified a pervasive cognitive tendency they termed the “belief in the law of small numbers”: people systematically treat small samples as though they are highly representative of the underlying population. This tendency is conventionally classified as a cognitive bias: a deviation from the normatively correct inference that small samples are unreliable [9, 10].

Within the present framework, this phenomenon is not an error but a structural consequence of Bayesian inference under strong priors. A Bayesian agent with informative prior distributions *should* make confident inferences from small samples—that is the entire computational advantage

of the Bayesian approach over the frequentist one. The “overconfidence” that Kahneman and Tversky documented is the subjective correlate of a system operating with priors strong enough to overwhelm small-sample likelihood. This is not a miscalibration; it is the only viable mode for a finite system that must act faster than frequentist convergence can deliver reliable estimates.

The distinction between “bias” and “informed prior” collapses once agent cost is taken into account. A system with $\mathcal{C}(a) \approx 0$ can afford to be an empiricist—to withhold judgment until the sample is large enough for the LLN to guarantee convergence. A system with $\mathcal{C}(a) \gg 1$ cannot afford this luxury; it must be a Bayesian—acting on prior-informed estimates that may be wrong in any given instance but are right often enough to sustain persistence. The error management perspective [11, 12] reinforces this: when loss functions are asymmetric, the Bayesian optimal decision rule is itself biased toward the less costly error, and this “bias” is a theorem of probability theory, not a deviation from it.

The survivorship principle completes the argument: we observe organisms with strong priors (organisms that act confidently from small samples) precisely because those are the organisms that acted quickly enough to persist. The “bias” is the fingerprint of the model-strategy, preserved by selection because the alternative—prior-free frequentist inference—is incompatible with persistence under conditions of irreplaceability.

9.4 The Will Paradox

The model-strategy contains an inherent paradox [26]: the mechanism that could correct structural distortion—volitional redirection of attention toward excluded domains—is, under default conditions, recruited to maintain the existing perceptual configuration. If the joint operation of cultural filters, moral constraints, and volitional regulation sets the prior weight for a domain to zero, no amount of evidence can shift the posterior. The only effective intervention is sustained reallocation of attentional duration toward excluded domains [26].

10 The Survivorship Principle

10.1 Persistence as Precondition, Not Goal

Definition 10.1 (Survivorship Principle). Persistence is a *precondition of observability*, not a property that systems possess by design or pursue by intention. We observe only those systems that persist; systems that fail to persist are not available for observation. The apparent “drive toward survival” is a selection artifact.

This operates at multiple levels. In biology, organisms appear designed for survival because those not so “designed” left no descendants [59]. In physics, stable configurations persist and unstable ones do not: a hydrogen atom persists not because it “wants to” but because its configuration is energetically stable. The principle generalizes: any system we observe has persisted long enough to enter our observational field.

10.2 The Parallel Structure

The survivorship principle reveals a structural parallel between the LLN and physical persistence, though the parallel must be stated carefully to avoid a conflation that would undermine it. In probability theory, the expectation $\mu = \int x dF(x)$ is a well-defined property of the distribution F , existing independently of any observations. The LLN tells us that the sample mean converges to this pre-existing quantity. The parallel with physical systems is therefore not that the “target” of persistence is constituted by the act of persisting (it is not), but rather that the *distribution F from which the system samples is itself a product of persistence*. The probability distribution governing a bacterium’s nutrient encounters is not a Platonic object given in advance; it is constituted by the ecosystem—itsself a collection of persisting systems—in which the bacterium

is embedded. If the ecosystem collapses, the distribution ceases to exist. In this sense, the “population parameter” toward which LLN convergence would proceed is itself contingent on a prior act of persistence at a higher organizational level.

- (i) In physics: stable configurations persist and unstable ones do not. A hydrogen atom persists not because it “wants to” but because its configuration is energetically stable.
- (ii) In biology: organisms do not survive because they “strive”; we observe only the survivors [59].
- (iii) In cognition: the brain does not seek truth; it maintains representations supporting persistence. We call these “knowledge” because representations that failed—and the organisms maintaining them—are no longer available for examination.
- (iv) In probability: the distribution F from which the LLN guarantees convergence is itself sustained by the persistence of the system that generates the observations.

10.3 Teleological Illusion as Predicted Distortion

The perception of teleology in non-teleological processes is itself predicted by the SDP [26]. A cognitive system modeling the world in terms of goals and intentions acts more effectively than one modeling it in terms of statistical residuals [10]. The teleological interpretation is computationally cheaper, making it exactly the kind of compression the model-strategy produces. The framework is thus self-applicable: the theory predicts the distortions that observers are likely to impose on the phenomena the theory describes.

11 The Hierarchical Nesting of Strategies

The preceding sections might create the impression that the LLN-strategy and the model-strategy are alternatives—that a system chooses one or the other based on its position on the replaceability continuum. This impression, while pedagogically useful, is misleading. The two strategies are not alternatives on a flat continuum; they are *hierarchically nested*. Every model-strategy is built upon a foundation of LLN-strategy operating at a different timescale.

11.1 The Training–Deployment Asymmetry

Consider the EPFL robot that spontaneously discovered gait transitions when trained for viability [60]. During training, the virtual robot fell billions of times. Each virtual instance was cheap, expendable, and perfectly replaceable— $\mathcal{R}(a) \approx 1$ and $\mathcal{C}(a) \approx 0$. The training process was pure LLN-strategy: massive population of trials, gradient descent as statistical averaging in parameter space, convergence guaranteed by the law of large numbers operating over the distribution of training episodes. But once the trained policy was deployed on a physical robot—expensive, singular, irreplaceable—it operated as a model-strategy: one chance per step, no time for re-sampling, rapid Bayesian inference from sensor data against learned priors.

The same asymmetry appears at every level of biological organization:

- (i) **Evolution.** Billions of organisms lived and died over hundreds of millions of years—each one a “training sample” in a population-level optimization. The LLN operated over this immense sample to produce, through natural selection, the cerebellar architecture that enables a cat to jump without calculation. Evolution is the LLN-strategy; the cerebellum is its compressed output.

- (ii) **Development.** A kitten misses hundreds of jumps before the cerebellar model is calibrated. Each developmental trial is relatively cheap (the kitten is protected by parental care). The LLN operates over the developmental sample; the adult cat’s motor competence is the compressed result.
- (iii) **Culture.** Thousands of generations of legal disputes, failed institutions, and social experiments constitute the “training data” for a civilization’s legal system. Each historical case is a sample; the body of precedent is the compressed model that enables a judge to decide a novel case without re-running the entire history of jurisprudence.

11.2 The Encapsulation Principle

This observation yields a structural principle that the paper’s original framework did not articulate:

Definition 11.1 (Encapsulation Principle). The model-strategy does not replace the LLN-strategy; it *encapsulates* it. Every internal model is a compressed encoding of statistical regularities accumulated through an LLN-process operating at a longer timescale or across a larger population. The two strategies are not alternative regimes but hierarchical levels of a single architecture.

Formally, let θ denote the parameters of an internal model (synaptic weights, cultural norms, institutional rules). These parameters were tuned by a process that satisfies LLN conditions at the *training timescale* τ_{train} : large sample size (many evolutionary trials, many developmental attempts, many historical precedents), approximate independence (each trial produces a noisy but roughly independent gradient signal), and convergent averaging (gradient descent, natural selection, precedent accumulation). Once the parameters are fixed, the model operates at the *deployment timescale* $\tau_{\text{deploy}} \ll \tau_{\text{train}}$, where none of these conditions hold: the agent faces a single situation, observations are dependent on the model itself, and there is no opportunity for re-sampling.

The model-strategy is thus the LLN-strategy passed through a compression bottleneck:

$$\underbrace{\text{Many cheap trials}}_{\text{LLN at } \tau_{\text{train}}} \xrightarrow{\text{compression}} \underbrace{\theta}_{\text{model parameters}} \xrightarrow{\text{deployment}} \underbrace{\text{One costly action.}}_{\text{Bayesian at } \tau_{\text{deploy}}} \quad (15)$$

The Kolmogorov complexity framework of Section 5 provides the formal content of this compression: θ is the shortest description of the regularities in the training data that are relevant to persistence at τ_{deploy} . The LLN guarantees that these regularities are accurately estimated (given sufficient training); the model encodes them in a form accessible for real-time Bayesian inference.

Hierarchical Encapsulation of Strategies

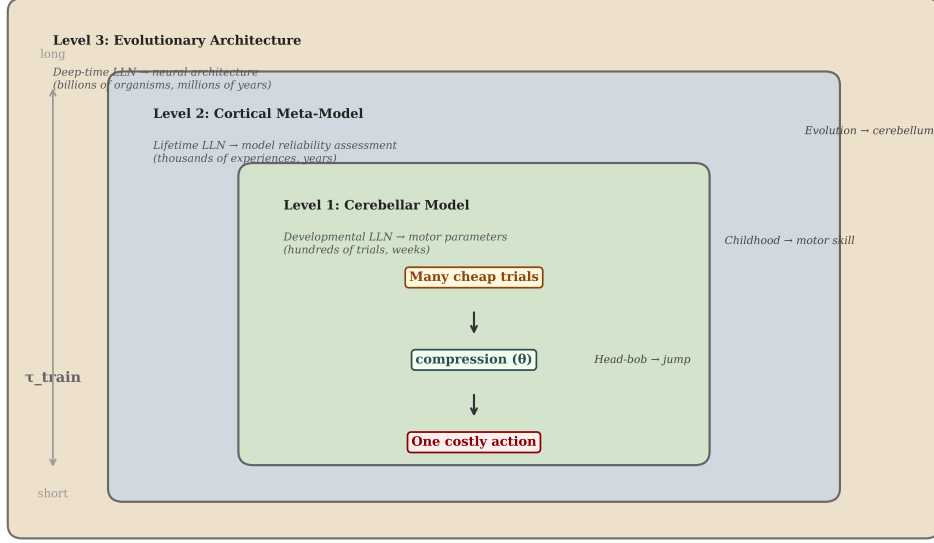


Figure 3: **Hierarchical encapsulation of strategies.** Each level contains a complete LLN→compression→action cycle operating at a different timescale. The innermost level (cerebellar model) operates on milliseconds; the middle level (cortical meta-model) on a lifetime; the outermost level (evolutionary architecture) on millions of years. The microgravity experiments of Section 8.7 demonstrate all three levels operating simultaneously when the model’s priors are invalidated.

11.3 Implications

This hierarchical nesting has three consequences for the framework:

First, there is no “pure” model-strategy. Every internal model rests on a statistical foundation—the only question is how many hierarchical levels of compression separate the current model from the raw statistical base. A bacterium’s temporal comparison circuit reflects billions of years of evolutionary selection; a human’s cultural intuitions reflect millennia of social experimentation; a neural-network robot’s policy reflects billions of simulated steps. The LLN never disappears; it recedes to a lower floor of the architecture.

This raises the question of *recursion*: if each model-strategy encapsulates an LLN-process, does the LLN-process at the training level itself rest on a model-strategy at a still longer timescale? In biological systems, the answer is yes: evolution’s “random search” is not truly random but is constrained by developmental biases, evolvability architectures, and the structure of mutation operators—all of which are themselves products of prior evolutionary selection. The recursion continues: the structure of the genetic code, the universality of the translation machinery, the conserved signaling pathways that bias the space of accessible phenotypes—each is a compressed model inherited from a deeper evolutionary past. Whether this recursion has a “bottom” is an open question; it may terminate at the boundary conditions set by physics and chemistry, or it may be the case that physical law itself represents a compression whose statistical foundation is inaccessible.

Second, the critical threshold $\mathcal{C}^*$ of Proposition 6.3 applies *within each timescale*. At the evolutionary timescale, agents (organisms) are cheap and the LLN operates freely. At the individual-action timescale, agents may be irreplaceable and the model-strategy is mandatory. The same system occupies different positions on Figure 1 depending on which timescale is considered.

Third, the quality of the model-strategy is bounded by the quality of the underlying LLN-

Cross-Domain Manifestation of the Strategy Transition			
Property	LLN-Strategy (High R)	Transitional	Model-Strategy (Low R)
Agent cost $c(a)$	Low	Moderate	High
Replaceability $R(a)$	≈ 1	Intermediate	$\rightarrow 0$
Selectivity σ	≈ 0	Emergent	High
Internal model	None	Temporal comparison	Predictive processing
Error tolerance	Individual expendable	Mixed	Individual critical
Biological example	Bacterial population	E. coli chemotaxis	Mammalian cognition
Social analogue	Market averaging	Polling / sampling	Expert judgment
Timescale	Generational	Seconds	Lifetime learning

Figure 4: **Cross-domain manifestation of the strategy transition.** Properties of systems at different points along the replaceability continuum.

process. A model trained on insufficient data, or on data from a distribution that has since shifted, will produce systematic errors—not the adaptive biases described in Section 9, but genuine maladaptation. This is the precise condition under which cognitive distortions cease to be optimal and become pathological: when the statistical foundation on which the model was trained no longer matches the environment in which it is deployed. The Structural Distortion Principle [26] predicts exactly this pattern: structural distortion is adaptive when priors match environmental regularities and maladaptive when they do not.

12 Cross-Domain Evidence and Applications

12.1 Biology: From Viruses to Vertebrates

Viral populations exemplify the extreme LLN-strategy: enormous population sizes ($N > 10^{10}$), high mutation rates, and perfectly replaceable virions. Insects represent an intermediate case with rudimentary modeling. Social insects distribute the modeling function across many individually expendable agents [61]. Vertebrates, particularly mammals, occupy the high-cost end. The mammalian brain—the most metabolically expensive organ per unit mass—is the physical substrate of the model-strategy, with predictive processing [16, 17] as its structural consequence.

12.2 Cognitive Science: Heuristics as Resource-Rational Strategies

The heuristics-and-biases program [9, 10] documented systematic deviations from normative rationality. The resource-rational analysis program [21] reinterpreted these as optimal given computational constraints. Recent comprehensive treatments of bounded rationality [62] further establish that the revolution initiated by Simon has become a foundational pillar of decision science. The present framework provides the structural foundation: normative rationality implicitly assumes the agent can “wait for convergence.” When agent cost is high, this assumption fails, producing heuristic shortcuts for extracting useful information from small samples—precisely the regime where the LLN provides no guarantees.

12.3 Social Systems: Institutions as Distributed Models

Social institutions [63] extend the model-strategy beyond individual agents. Legal systems maintain collective models (bodies of law) encoding accumulated precedent. The market-as-averaging-mechanism (efficient market hypothesis) represents an attempt to apply the LLN-strategy to economics; its systematic failures—bubbles, crashes, mispricings [23, 27, 30]—are predicted by the present framework since economic agents violate independence (herding), stationarity (regime changes), and finite variance (heavy tails).

12.4 The Conant-Ashby Theorem

The Conant-Ashby theorem [61]—that every good regulator must contain a model of its system—is a special case of the present framework. Its conditions (viable performance) correspond to agent cost exceeding $\mathcal{C}^*$. Below this threshold, regulation is unnecessary; above it, modeling is required.

13 Integrated Architecture

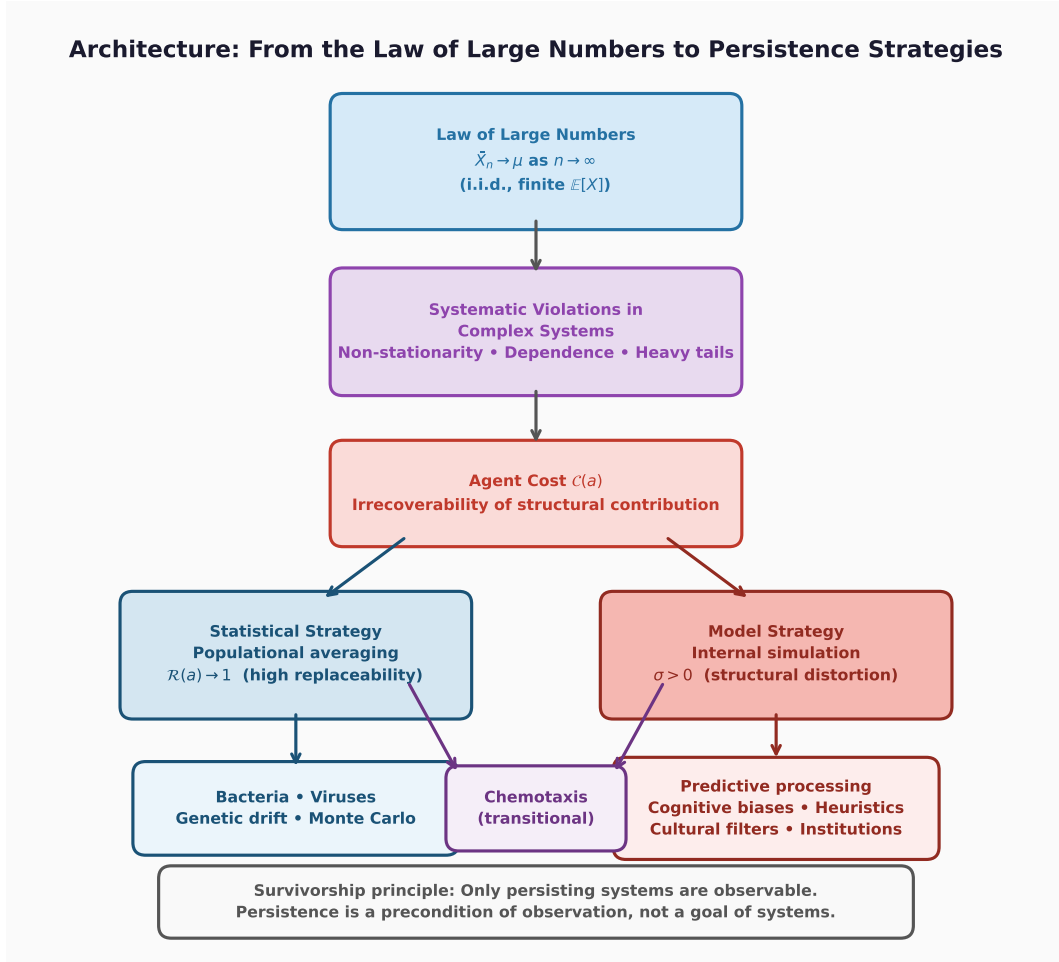


Figure 5: **Integrated architecture.** The LLN’s formal conditions (top) are systematically violated in complex systems. Agent cost determines strategy adoption. The survivorship principle (bottom) provides the meta-level observation that the entire architecture is visible only because we observe persisting systems.

14 Testable Predictions

- P1. Replaceability–selectivity correlation.** Species with lower individual replaceability should exhibit higher representational selectivity, measurable as the ratio of attended to available environmental information.
- P2. Chemotaxis threshold.** Organisms in the transitional regime should exhibit proto-modeling behaviors whose complexity scales with agent cost. The temporal integration window in chemotactic navigation should correlate with individual generation time across microbial species.
- P3. Institutional failure under LLN-violation.** Institutions designed on LLN-assumptions (insurance, market averaging, polling) should fail systematically when conditions shift toward dependence, non-stationarity, or heavy tails. The failure mode should be predictable from which LLN condition is violated.
- P4. Bias optimality under cost manipulation.** Reducing the perceived cost of error should reduce measured biases. Increasing perceived irreversibility should amplify biases, controlling for emotional arousal.
- P5. Attentional redistribution without information addition.** Following [26], sustained reallocation of attentional duration toward previously excluded domains should produce measurable changes in perceptual sensitivity even when no new information is provided.
- P6. Maladaptation from encapsulation failure.** The encapsulation principle (Section 11) predicts that cognitive distortions become maladaptive—not merely imprecise but systematically counter-productive—when the statistical foundation on which the internal model was trained no longer matches the deployment environment. This should be observable as a specific pattern: organisms or institutions displaced into novel environments should exhibit biases that were adaptive in the ancestral environment but are harmful in the new one, and the degree of maladaptation should correlate with the magnitude of distributional shift between training and deployment environments. This prediction distinguishes the present framework from accounts that treat all biases as either universally adaptive or universally irrational.

15 Relation to the Broader Research Program

The present paper addresses a specific gap within a broader research program developed across seventy-five publications spanning seven domains over twenty-seven years [24, 26, 34]. The program is organized around a central thesis: *persistence, not truth, is the primary organizing principle of complex systems*. The dynamical core proves that persisting systems must satisfy closure, boundary maintenance, and resilience conditions, and that four laws of self-organization are independent and complete. The cognitive domain demonstrates that perceptual distortion is structural rather than accidental, and that predictive processing is derivable from physical constraints.

The present paper fills a gap that this body of work revealed: *what determines which persistence strategy a system adopts*. The concept of agent replaceability, the encapsulation principle, and the Kolmogorov compressibility criterion provide this bridge between the thermodynamic results (which establish the costs of persistence) and the cognitive results (which characterize the architecture of bounded modeling).

16 Discussion

16.1 Summary of Contributions

This paper establishes four principal results. First, the conditions required by the LLN are systematically violated by complex adaptive systems, and these violations are structurally entailed by the requirements of persistence; moreover, intermediate convergence theorems (ergodic, martingale, mixing) do not resolve the problem for systems with high agent cost. Second, the transition from statistical averaging to internal modeling is governed by agent replaceability, with a complementary characterization in terms of Kolmogorov complexity: the transition occurs when the environment contains compressible structure at the agent’s operational scale that averaging would destroy; the agent-cost criterion determines *whether* a system must model, while the compressibility criterion determines *whether* modeling is possible. Third, cognitive biases and cultural filtering are structural consequences of operating in the model-strategy regime—optimal in the non-circular sense of exploiting scale-specific compressible regularities—and the apparent teleology of complex systems is a survivorship artifact. Fourth, the encapsulation principle establishes that the two strategies are not alternatives but hierarchical levels: every internal model is a compressed encoding of statistical regularities accumulated through an LLN-process at a longer timescale, and this nesting may be recursive.

16.2 Relation to Existing Literature

The framework provides the structural foundation for the resource-rational analysis program [21] by identifying agent cost as the parameter determining when biased strategies are favored. It extends the Conant-Ashby theorem [61] by specifying when modeling becomes necessary versus unnecessary. It connects non-equilibrium thermodynamics [13, 14] to cognitive science [16, 17, 20] through structural transformation costs.

16.3 Limitations

The framework is primarily theoretical; the testable predictions require empirical investigation. The sigmoid transition between strategies is a qualitative prediction; its precise functional form is likely system-dependent. The survivorship principle concerns interpretation rather than content of observations, limiting direct empirical test.

16.4 Future Directions

Computational implementation of the strategy transition via agent-based models would enable precise characterization of the transition boundary. Comparative neuroscience studies measuring sensory filtering ratios across species would test Prediction P1. Analysis of institutional failures through the lens of LLN-condition violations would test Prediction P3 and inform institutional design.

17 Conclusion

The law of large numbers guarantees convergence of averages to truth—given enough independent observations from the same distribution. But the conditions that make this convergence possible are precisely those that complex adaptive systems cannot satisfy. Their observations are dependent, their environments non-stationary, and their risks heavy-tailed.

When the LLN fails, systems face a choice determined by structure, not intention. Systems of cheap, replaceable agents persist through statistical mass—they are, in effect, Monte Carlo methods implemented in biological substrate. Systems of expensive, irreplaceable agents must develop internal models—Bayesian inference engines with strong priors, shaped by evolution,

development, and culture—that enable action from sparse data without waiting for frequentist convergence.

The cognitive biases that the model-strategy produces are not failures of rationality but theorems of Bayesian decision theory applied under asymmetric loss functions and strong priors. The cultural filters shaping perception are not obstacles to truth but necessary compressions that encode environmental regularities as priors, making action possible in environments too rich for any finite system to sample exhaustively. The apparent drive toward survival pervading living systems is not a mysterious force but a tautology: we see survival because we can only see what survives.

The framework locates the deepest division in the organization of complex systems not in the distinction between living and non-living, or between simple and complex, but in a mathematical property: *whether the system can wait for the law of large numbers to take effect, or whether it must act before convergence occurs*. On one side lie populations, statistical ensembles, and randomized algorithms. On the other lie organisms, minds, institutions, and civilizations—all operating in the shadow of a mathematical guarantee that does not apply to them at the timescale of action, all improvising with Bayesian models whose priors were forged by the very law of large numbers that no longer applies, and all persisting not because they are optimal but because the alternatives have already disappeared.

The deepest insight may be this: the law of large numbers is not a strategy that complex systems abandon. It is a strategy they *hide inside themselves*. The cat’s cerebellum is the law of large numbers, compressed through evolution and learning into neural circuits. The robot’s policy network is the law of large numbers, compressed through simulation into weights. The judge’s legal intuition is the law of large numbers, compressed through millennia of precedent into institutional memory. Nowhere in this hierarchy does statistical convergence disappear—it merely descends to a lower floor, invisible from above but structurally indispensable.

References

- [1] Jacob Bernoulli. *Ars conjectandi*. 1713.
- [2] William Feller. *An Introduction to Probability Theory and Its Applications*, volume 1. John Wiley & Sons, 3 edition, 1968.
- [3] Patrick Billingsley. *Probability and Measure*. John Wiley & Sons, 3 edition, 1995.
- [4] Nicholas Metropolis and Stanislaw Ulam. The Monte Carlo method. *Journal of the American Statistical Association*, 44(247):335–341, 1949.
- [5] A. N. Kolmogorov. Sur la loi forte des grands nombres. *Comptes Rendus de l’Académie des Sciences*, 191:910–912, 1930.
- [6] A. Ya. Khinchin. Sur la loi des grands nombres. *Comptes Rendus de l’Académie des Sciences*, 188:477–479, 1929.
- [7] Nasrollah Etemadi. An elementary proof of the strong law of large numbers. *Zeitschrift für Wahrscheinlichkeitstheorie und verwandte Gebiete*, 55:119–122, 1981.
- [8] Rick Durrett. *Probability: Theory and Examples*. Cambridge University Press, 5 edition, 2019.
- [9] Amos Tversky and Daniel Kahneman. Judgment under uncertainty: Heuristics and biases. *Science*, 185(4157):1124–1131, 1974.
- [10] Daniel Kahneman. *Thinking, Fast and Slow*. Farrar, Straus and Giroux, 2011.

- [11] Martie G. Haselton and David M. Buss. Error management theory: A new perspective on biases in cross-sex mind reading. *Journal of Personality and Social Psychology*, 78(1):81–91, 2000.
- [12] Kevin R. Foster and Hanna Kokko. The evolution of superstitious and superstition-like behaviour. *Proceedings of the Royal Society B*, 276(1654):31–37, 2009.
- [13] Ilya Prigogine and Isabelle Stengers. *Order Out of Chaos: Man’s New Dialogue with Nature*. Bantam, 1984.
- [14] Jeremy L. England. Statistical physics of self-replication. *The Journal of Chemical Physics*, 139(12):121923, 2013.
- [15] Christopher Jarzynski. Nonequilibrium equality for free energy differences. *Physical Review Letters*, 78(14):2690–2693, 1997.
- [16] Karl Friston. The free-energy principle: A unified brain theory? *Nature Reviews Neuroscience*, 11(2):127–138, 2010.
- [17] Andy Clark. Whatever next? predictive brains, situated agents, and the future of cognitive science. *Behavioral and Brain Sciences*, 36(3):181–204, 2013.
- [18] Jakob Hohwy. The predictive mind. 2013.
- [19] Ryan Smith, Paul Badcock, and Karl J. Friston. Recent advances in the application of predictive coding and active inference models within clinical neuroscience. *Psychiatry and Clinical Neurosciences*, 75(1):3–13, 2021.
- [20] Rowan Hodson, Manesh Bhatt, Sukhwinder S. Bhatt, and Ryan Smith. Empirical evidence for predictive processing: A systematic review. *Neuroscience & Biobehavioral Reviews*, 157:105538, 2024.
- [21] Falk Lieder and Thomas L. Griffiths. Resource-rational analysis: Understanding human cognition as the optimal use of limited computational resources. *Behavioral and Brain Sciences*, 43:e1, 2020.
- [22] Andrei A. Markov. Extension of the law of large numbers to dependent quantities. *Izvestiya Fiziko-Matematicheskogo Obshchestva pri Kazanskom Universitete*, 15:135–156, 1906.
- [23] Benoit Mandelbrot. The variation of certain speculative prices. *The Journal of Business*, 36(4):394–419, 1963.
- [24] Boris Kriger. The transformational basis of persistence: A formal theory of structural viability. *Zenodo*, 2026. doi: 10.5281/zenodo.18435982.
- [25] Raymond L. Lindeman. The trophic-dynamic aspect of ecology. *Ecology*, 23(4):399–417, 1942.
- [26] Boris Kriger. The structural distortion principle: A closed-loop model of perception, attention, and world-maintenance in bounded cognitive systems. *Zenodo*, 2026. doi: 10.5281/zenodo.18452700.
- [27] Nassim Nicholas Taleb. *The Black Swan: The Impact of the Highly Improbable*. Random House, 2007.
- [28] Nassim Nicholas Taleb, Yaneer Bar-Yam, and Pasquale Cirillo. On single point forecasts for fat-tailed variables. *International Journal of Forecasting*, 38(2):413–422, 2022.

- [29] Pasquale Cirillo and Nassim Nicholas Taleb. Tail risk of contagious diseases. *Nature Physics*, 16:606–613, 2020.
- [30] Nassim Nicholas Taleb. *Statistical Consequences of Fat Tails: Real World Preasymptotics, Epistemology, and Applications*. STEM Academic Press, 2023.
- [31] George D. Birkhoff. Proof of the ergodic theorem. *Proceedings of the National Academy of Sciences*, 17(12):656–660, 1931.
- [32] Joseph L. Doob. Stochastic processes. 1953.
- [33] Richard C. Bradley. *Introduction to Strong Mixing Conditions*. Kendrick Press, 2005.
- [34] Boris Kriger. Universal laws of self-organization: Proven independent and complete. *Zenodo*, 2026. doi: 10.5281/zenodo.18363729.
- [35] A. N. Kolmogorov. *Three approaches to the quantitative definition of information*, volume 1. 1965.
- [36] Ming Li and Paul Vitányi. *An Introduction to Kolmogorov Complexity and Its Applications*. Springer, 3 edition, 2008.
- [37] Peter D. Grünwald. *The Minimum Description Length Principle*. MIT Press, 2007.
- [38] Daniel M. Wolpert and Zoubin Ghahramani. Computational principles of movement neuroscience. *Nature Neuroscience*, 3:1212–1217, 2000.
- [39] Rolf Landauer. Irreversibility and heat generation in the computing process. *IBM Journal of Research and Development*, 5(3):183–191, 1961.
- [40] Antoine Bérut, Artak Arakelyan, Artyom Petrosyan, Sergio Ciliberto, Raoul Dillenschneider, and Eric Lutz. Experimental verification of Landauer’s principle linking information and thermodynamics. *Nature*, 483:187–189, 2012.
- [41] Julia J. Harris, Renaud Jolivet, and Upinder S. Bhalla. Synaptic energy use and supply. *Neuron*, 75(5):762–777, 2012.
- [42] Claude E. Shannon. A mathematical theory of communication. *The Bell System Technical Journal*, 27(3):379–423, 1948.
- [43] Thomas M. Cover and Joy A. Thomas. *Elements of Information Theory*. John Wiley & Sons, 2 edition, 2006.
- [44] Howard C. Berg and Douglas A. Brown. Chemotaxis in Escherichia coli analysed by three-dimensional tracking. *Nature*, 239:500–504, 1972.
- [45] Howard C. Berg. E. coli in motion. 2004.
- [46] Julius Adler. Chemotaxis in bacteria. *Science*, 153(3737):708–716, 1966.
- [47] Biyi Zhou, Christine M. Szymanski, and Arden Bhojlan. Bacterial chemotaxis in human diseases. *Trends in Microbiology*, 31(5):453–467, 2023.
- [48] George H. Wadhams and Judith P. Armitage. Making sense of it all: Bacterial chemotaxis. *Nature Reviews Molecular Cell Biology*, 5:1024–1037, 2004.
- [49] Victor Sourjik and Ned S. Wingreen. Responding to chemical gradients: Bacterial chemotaxis. *Current Opinion in Cell Biology*, 24(2):262–268, 2012.

- [50] Takahiro Miki, Joonho Lee, Jemin Hwangbo, Lorenz Wellhausen, Vladlen Koltun, and Marco Hutter. Learning robust perceptive locomotion for quadrupedal robots in the wild. *Science Robotics*, 7(62):eabk2822, 2022.
- [51] Ilija Radosavovic, Tete Xiao, Bike Zhang, Trevor Darrell, Jitendra Malik, and Koushil Sreenath. Real-world humanoid locomotion with reinforcement learning. *Science Robotics*, 9(89):eadi9579, 2024.
- [52] Milad Shafiee, Guillaume Bellegarda, and Auke Ijspeert. Viability leads to the emergence of gait transitions in learning agile quadrupedal locomotion on challenging terrains. *Nature Communications*, 15:3073, 2024.
- [53] Siegfried J. Gerathewohl and Herbert D. Stallings. The labyrinthine posture reflex (righting reflex) in the cat during weightlessness. *The Journal of Aviation Medicine*, 28(4):345–355, 1957.
- [54] Shigeo Mori. Disorientation of animals in microgravity. *Nagoya Journal of Medical Science*, 58:71–81, 1995.
- [55] USAF Aerospace Medical Research Laboratories. Weightless man: Self-rotation techniques. Technical report, Behavioral Sciences Laboratory, Wright-Patterson Air Force Base, 1962.
- [56] Thomas R. Kane and M. P. Scher. A dynamical explanation of the falling cat phenomenon. *International Journal of Solids and Structures*, 5(7):663–670, 1969.
- [57] Raymond S. Nickerson. Confirmation bias: A ubiquitous phenomenon in many guises. *Review of General Psychology*, 2(2):175–220, 1998.
- [58] Amos Tversky and Daniel Kahneman. Belief in the law of small numbers. *Psychological Bulletin*, 76(2):105–110, 1971.
- [59] Richard Dawkins. *The Selfish Gene*. Oxford University Press, 1976.
- [60] Moritz Vollenweider, Marko Bjelonic, Victor Klemm, Nikita Rudin, Joonho Lee, and Marco Hutter. Spontaneous emergence of gaits in quadrupedal robots trained for viability. *Science Robotics*, 2024.
- [61] Roger C. Conant and W. Ross Ashby. Every good regulator of a system must be a model of that system. *International Journal of Systems Science*, 1(2):89–97, 1970.
- [62] Sanjit Dhami and Cass R. Sunstein. *Bounded Rationality: Heuristics, Judgment, and Public Policy*. MIT Press, 2022.
- [63] W. Ross Ashby. An introduction to cybernetics. 1956.